The Electron and the Planck Constant in the Rope Framework

An Exactly-Solved Electron Profile, a Size Prediction, and What Treating ħ as Constant Assumes

A Technical Report of the Rope Framework Programme

Mark Palmer · with computational collaboration by Claude (Anthropic) · palmer100@gmail.com

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this corpus develops and formalises it. · *August 2026*

This report presents the standing, positive results of the electron-structure campaign — what the framework now knows about the electron and about the Planck constant. The results are conditional (Modeled); the extensive candidate-elimination behind them is left to the registry.

Abstract

What is the electron, mechanically, and where does the Planck constant ħ come from? This report presents the standing results of the Rope Framework's electron-structure campaign, stated as successes rather than as the long search that produced them. Four results define the picture. First, the electron's three-dimensional profile is *solved exactly*: the governing equation turns out to be exactly soluble, yielding a cusped, scale-free family with finite energy. Second, the framework *selects its own electron functional* — the pure-tension form is forced by the framework's own denial of material points, not chosen. Third, the electron's size comes out near **0.407 fm**, a value that survives a genuinely different estimator to 1.3%. Fourth, on ħ: no combination of the medium's constants yields ħ (it is a mesoscopic scale, tens of mesh-spacings across), and ħ's constancy reduces *exactly* to the constancy of one product, tension times area-per-strand. This is the electron whose dressing enters the fine-structure chain of the companion alpha paper; here its structure and scale are pinned down and the ħ question is stated precisely.

1. The Electron's Profile, Solved Exactly

The electron in the framework is a terminated winding in the strand medium — a knot-like defect where the two-strand rope ends. The central structural question is its radial profile: how the winding's intensity falls off from the core outward. For several sessions this was approached with scaling arguments and an assumed profile shape. The campaign's turning point was the discovery that the profile does not need to be assumed: the governing Euler-Lagrange equation is **exactly soluble**.

The reason is structural. The energy functional carries no non-derivative term, which makes a certain combination a first integral of the motion; the resulting function is strictly increasing, so its inversion is unique, and the entire family of solutions collapses to one universal profile. That profile is *cusped and scale-free*, with finite total energy — and it refuted the smooth (sech-like) shape the earlier sessions had assumed. Solving the equation rather than guessing it is what converted the electron's structure from a modelling choice into a derived form, and it reduced the remaining freedom to a single number: the overall scale.

2. The Framework Selects Its Own Electron

A subtlety surfaced once the profile was in hand: two different energy functionals had been carried through the work as though they were one, and they are not equivalent. Which is the framework's? The answer came from a derived commitment rather than from taste, and it is a clean example of the framework constraining itself.

The framework denies material points — the medium is strands, with nothing pointlike to displace. A medium with no material points has nothing to move longitudinally, so an elastic energy in the longitudinal displacement is simply *not available to it*; minimising over that displacement is minimising over a mere labelling, not over physics. What remains is the strand's curve and its length — the **pure-tension functional**. The framework's own ontology therefore selects the pure-tension form and supersedes the alternative. This is the kind of result the programme values most: a modelling fork closed by a derived principle the framework already held, not by a fitted preference.

3. The Electron's Size, and a Scale That Is Exactly Invariant

With the profile solved and the functional selected, the electron's size becomes computable. The campaign's standing value is near **0.407 fm**, and its credibility rests on a cross-check rather than a single computation: two estimators that fail in genuinely different ways — a discrete trapezoidal integration with a truncation rule, versus an analytic integral of a model fit — agree to 1.3%. Because the two methods have unrelated weaknesses, their agreement is worth more than either alone; the natural objection to the first (that its truncation was a choice) does not touch the second.

Underpinning the size is a scale relation that came out *exactly invariant*: the strand spacing satisfies $w = a/\sqrt{3}$, where a is the mesh scale. This was initially registered as a choice between competing referents, but under the framework's decide-before-computing discipline it turned out not to be a choice at all — identifying strand count with energy density is a two-line consequence of additivity (each strand carries the tension $T0$, so strands-per-area is energy-density over $T0$, and energy density is the field squared). There was never a second referent; the branch is forced, and the spacing is exact.

4. The Planck Constant: What Its Constancy Assumes

The deepest result of the arc is also its most honest, and it concerns the Planck constant — the one quantum input the framework had hoped to derive from the medium. The campaign's finding is a precise structural statement, and it is worth stating exactly because it reframes the question rather than papering over it.

ħ is not hidden in the medium's constants. A dimensional audit is decisive: from the medium's two lengths, one tension, and one speed, *no combination lands within three orders of magnitude of ħ*. ħ is not a rearrangement of the framework's primitives. Instead it is a *mesoscopic* scale — an action associated with a patch of medium roughly seventy-five mesh-spacings across — which is why no local combination of constants produces it.

ħ's constancy reduces to one product. The positive content is sharp: the constancy of ħ across space reduces *exactly* to the constancy of $T \cdot w^2$ — tension times area-per-strand. The framework can now say precisely what it is assuming when it treats ħ as a universal constant: it is assuming the vacuum is homogeneous in that one product. This does not derive ħ, and the report does not claim it does; what it delivers is the exact condition ħ's universality depends on, which is a real advance over treating ħ as an unexamined input — and it points to where the framework's first sub-quantum-scale prediction would live.

5. Connection to the Fine-Structure Chain

This electron is not a separate object from the one in the companion alpha paper — it is the same terminated winding. The electron-dressing solver that computes the number D_E at the heart of the fine-structure chain ($1/\alpha = 4\pi^3$ D_E) operates on precisely the profile solved here. The structure this report pins down — the exact profile, the selected functional, the invariant spacing — is the foundation the alpha result stands on. The two reports are two views of one electron: this one describes what the electron *is* and how big it is; the alpha paper describes what its dressing *does* to the electromagnetic coupling.

6. Status

Every result here is labelled Modeled: conditional, with premises named. The table summarises what stands and what each result still leans on.

Result	What stands	What it is conditional on
Exact profile	The Euler-Lagrange equation is exactly soluble; one universal cusped, finite-energy family.	The energy functional's form (selected in §2).
Selected functional	Pure-tension form forced by the no-material-points commitment.	The framework's ontology holding as stated.
Size ≈ 0.407 fm	Survives two independent estimators to 1.3%.	The scale calibration; a ~28% cross-sector tension noted in the registry, not an artifact.
Invariant spacing $w = a/\sqrt{3}$	Forced by additivity; exact.	— (derived)
ħ structure	ħ is mesoscopic; its constancy = constancy of $T \cdot w^2$, exactly.	Does not derive ħ's value; states the homogeneity condition it assumes.

The through-line: the electron's structure is now derived where it was once assumed — an exact profile, a functional the framework selects for itself, an exactly-invariant spacing — and the ħ question has been sharpened from "where does it come from" to the exact product whose constancy it requires. These are the standing results; the search that found them, including the many candidates ruled out, lives in the claims registry for those who want the full route.

Provenance: registered as claims in the ELEC and HBAR families in claims.yaml, with per-session bars and benchmarks in analysis/ and benchmarks/. Companion to the alpha paper (ALPHA-DE-CHAIN) and the mass paper.